Q1.

It is given that the general solution of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$$

is $y = e^x(Ax + B)$. Hence find the general solution of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 6e^x$$

(Total 5 marks)

Q2.

Find the general solution of the differential equation

$$\frac{d^2y}{dt^2} - 6\frac{dy}{dt} + 10y = e^{2t}$$

giving your answer in the form $y = f(t)$.

(Total 6 marks)

Q3.

(a) Find the values of the constants a, b and c for which $a + bx + cxe^{-3x}$ is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = 3x - 8e^{-3x}$$

(5)

(b) Hence find the general solution of this differential equation.

(3)

(c) Hence express y in terms of x, given that $y = 1$ when $x = 0$ and that $\frac{dy}{dx} \to -1$ as $x \to \infty$.

(4)

(Total 12 marks)

Q4.

Solve the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 10y = 26e^x$$

given that $y = 5$ and $\frac{dy}{dx} = 11$ when $x = 0$. Give your answer in the form $y = f(x)$.

(Total 10 marks)

Q5.

It is given that, for $x \neq 0$, y satisfies the differential equation

$$x\frac{d^2y}{dx^2} + 2(3x + 1)\frac{dy}{dx} + 3y(3x + 2) = 18x$$

(a) Show that the substitution $u = xy$ transforms this differential equation into

$$\frac{d^2u}{dx^2} + 6\frac{du}{dx} + 9u = 18x$$

(4)

(b) Hence find the general solution of the differential equation

$$x\frac{d^2y}{dx^2} + 2(3x + 1)\frac{dy}{dx} + 3y(3x + 2) = 18x$$

giving your answer in the form $y = f(x)$.

(8)
(Total 12 marks)

Q6.

(a) Find the values of the constants p and q for which $p + qxe^{-2x}$ is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 4 - 9e^{-2x}$$

(5)

(b) Hence find the general solution of this differential equation.

(3)

(c) Hence express y in terms of x, given that $y = 4$ when $x = 0$ and that $\frac{dy}{dx} \to 0$ as $x \to \infty$.

(4)
(Total 12 marks)

Q7.

It is given that y satisfies the differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 2e^{-2x}$$

(a) Find the value of the constant p for which $y = pxe^{-2x}$ is a particular integral of the given differential equation.

(4)

(b) Solve the differential equation, expressing y in terms of x, given that $y = 2$ and

$\frac{dy}{dx}$ = 0 when x = 0.

(8)
(Total 12 marks)

Q8.

Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4y = 8x^2 + 9\sin x$$

(Total 8 marks)

Q9.

(a) Find the value of the constant k for which k sin $2x$ is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} + y = \sin 2x$$

(3)

(b) Hence find the general solution of this differential equation.

(4)
(Total 7 marks)

Q10.

It is given that y satisfies the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 8\sin x + 4\cos x$$

(a) Find the value of the constant k for which $y = k \sin x$ is a particular integral of the given differential equation.

(3)

(b) Solve the differential equation, expressing y in terms of x, given that y = 1 and $\frac{dy}{dx} = 4$ when x = 0.

(8)
(Total 11 marks)

Q11.

(a) Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 5y = 5$$

(6)

(b) Hence express y in terms of x, given that $y = 2$ and $\frac{dy}{dx} = 3$ when $x = 0$.

(4)
(Total 10 marks)

Q12.

(a) Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 10e^{-2x} - 9$$

(10)

(b) Hence express y in terms of x, given that $y = 7$ when $x = 0$ and that $\frac{dy}{dx} \to 0$ as $x \to \infty$.

(4)
(Total 14 marks)

Q13.

Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = 6 + 5\sin x$$

(Total 12 marks)

Q14.

(a) Find the value of the constant k for which kx^2e^{5x} is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 6e^{5x}$$

(6)

(b) Hence find the general solution of this differential equation.

(4)
(Total 10 marks)

Q15.

(a) Find the roots of the equation $m^2 + 2m + 2 = 0$ in the form $\quad + \mathrm{i}b$.

(2)

(b) (i) Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 4x$$

(6)

(ii) Hence express y in terms of x, given that $y = 1$ and $\frac{dy}{dx} = 2$ when $x = 0$.

(4)
(Total 12 marks)

Q16.

It is given that y satisfies the differential equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = 8x - 10\cos 2x$$

(a) Show that $y = 2x + \sin 2x$ is a particular integral of the given differential equation.

(3)

(b) Find the general solution of the differential equation.

(4)

(c) Hence express y in terms of x, given that $y = 2$ and $\frac{dy}{dx} = 0$ when $x = 0$.

(4)
(Total 11 marks)

Q17.

In this question, take $g = 10$ m s^{-2}.

A bungee jumper has mass 75 kg. He uses an elastic rope of natural length 12 m and modulus of elasticity 450 N. He falls vertically, and when the rope becomes taut for the first time, he is travelling at 12.5 m s^{-1}. Assume that, once the rope is taut, the bungee jumper experiences an air resistance force that has magnitude $15v$ newtons, where v m s^{-1} is his speed.

At time t seconds after the rope has become taut, the extension of the rope is x metres.

(a) Show that

$$10\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = 100$$

(4)

(b) Find x in terms of t.

(10)

(c) Find the value of t when the bungee jumper first comes instantaneously to rest.

(5)
(Total 19 marks)

Q18.

A particle P, of mass 0.4 kg, is attached to one end of a light elastic string, and the other end of the string is attached to a fixed point A.

(a) The particle P hangs in equilibrium at a point E, vertically below A, where the extension of the string is 0.2 metres. Calculate the stiffness of the string.

(3)

(b) The particle P is pulled vertically downwards from the point E by a distance of 0.1 metres, and released from rest. The displacement of P from E at time t seconds after being released is x metres.

(i) Given that the string does not become slack during the subsequent motion, show that

$$\ddot{x} = hx$$

where h is a constant to be determined.

(4)

(ii) Hence deduce that the motion of P is simple harmonic.

(1)

(iii) Show that the period of this motion is 0.898 seconds, correct to three significant figures.

(2)

(iv) Calculate the maximum speed of P during its motion.

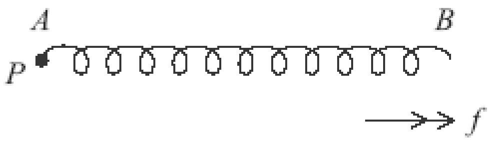
(2)

(Total 12 marks)

Q19.

A light spring AB lies at rest and unstretched on a smooth horizontal surface. A particle, P, of mass m is attached to the end A of the spring.

The spring has stiffness $2mn^2$, where n is a positive constant. The end B of the spring is set in motion and moves with constant acceleration of magnitude f in the direction AB, as shown in the diagram. The particle P is consequently forced into motion.



At time t after the motion begins, P is moving with speed v and experiences a resistant force of magnitude 3 mnv. The extension of the spring is x and the displacement of P from its initial position in the direction AB is y.

(a) Show that

$$y = \frac{1}{2}ft^2 - x$$

(3)

(b) Hence show that P has velocity

$$ft - \frac{dx}{dt}$$

(1)

(c) Hence show that

$$\frac{d^2x}{dt^2} + 3n\frac{dx}{dt} + 2n^2x = f + 3nft$$

(5)

(d) Given that n = 1, and that a particular integral for this differential equation is

$$x = \frac{f}{4}(6t - 7)$$

find an expression for x, in terms of f and t.

(7)
(Total 16 marks)

Q20.

A particle, of mass 2 kg, is suspended from a fixed point O by a light spring of natural length 0.5 metres and modulus of elasticity 49 N.

(a) Initially, the particle hangs at rest in equilibrium below O. Find the extension of the spring in this position.

(2)

(b) A force, F newtons, is then applied to the particle in a vertically downwards direction. The displacement of the particle below its equilibrium position at time t seconds later is x metres. Given that F = 12 cos nt, where n is a positive constant, show that

$$\frac{d^2x}{dt^2} + 49x = 6\cos nt$$

(5)

(c) In the case where n = 5, find an expression for x at time t.

(10)

(d) State the value of n for which resonance occurs.

(1)
(Total 18 marks)

Q21.

A particle P, of mass $2m$, is suspended from a fixed point O by a light elastic string of natural length a and modulus of elasticity $8mn^2a$, where n is a positive constant. The particle is released from rest when t = 0 at a point A, where A is vertically below O and $OA = a$.

When the particle is moving with speed v, it experiences air resistance of magnitude $4mnv$.

(a) The displacement of P below A at time t is x. Show that x satisfies the equation

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 4n^2x = g$$

(3)

(b) Find x in terms of n, g and t.

(12)

(c) Hence show that the first time the particle comes to rest is when $t = \frac{\pi}{\sqrt{3}n}$.

(4)

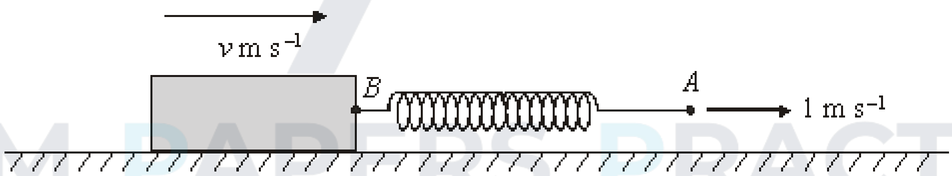
(Total 19 marks)

Q22.

A block of mass 0.5 kg moves on a smooth horizontal surface. It is attached to the end B of a light elastic spring AB, where AB is horizontal.

The natural length of the spring is 0.5 m and its stiffness is 5 N m^{-1}.

The end A of the spring is pulled horizontally with constant speed of 1 m s^{-1}, causing the block to move across the horizontal surface.



At time t seconds, the length of the spring is x metres and the block is moving with speed v m s^{-1}.

The motion of the block is subject to a resistive force of v N.

(a) Write down an expression for the tension in the spring in terms of x.

(2)

(b) Show that the equation of motion of the block is

$$\frac{dv}{dt} + 2v - 10x = -5.$$

(3)

(c) (i) Explain why $\dot{x} = 1 - v$.

(1)

(ii) Hence show that

$$\ddot{x} + 2\dot{x} + 10x = 7.$$

(2)

(d) Find the general solution of this equation.

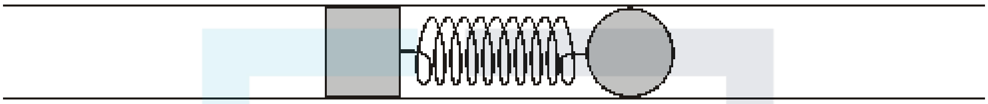
(6)

(e) State the limiting value of x. Give a reason for your answer.

(2)
(Total 16 marks)

Q23.

A solid ball P is attached to one end of a light elastic spring. The other end of the spring is attached to a piston S which is free to move horizontally inside a smooth tube, as shown in the diagram.



The mass of P is 0.5 kg. The spring has natural length 0.5 m and stiffness 4.5 N m^{-1}. The system initially rests in equilibrium.

The piston S is forced to oscillate, causing P to move. At time t seconds, the displacements of S and P are $\frac{1}{2}$ sin 2t metres and x metres respectively, measured in the same direction from their initial positions.

(a) Draw a diagram showing the forces acting on P and the displacements of P and S.

(1)

(b) Show that x satisfies the differential equation

$$\ddot{x} + 9x = \frac{9}{2}\sin 2t.$$

(4)

(c) Find x as a function of t.

(9)

(d) Show that P is instantaneously at rest when $t = \frac{2\pi}{5}$.

(2)
(Total 16 marks)

Q24.

During the design of an aircraft, part of its tail section is subjected to vibrations. At time t, the displacement of the tail section, x, is modelled by the differential equation

$$\ddot{x} + 121x = 3 \sin \omega t.$$

Different values of ω are used to simulate different flight conditions.

Find the general solution of this differential equation when $\omega \neq 11$.

(Total 6 marks)

Q25.

The pointer on a set of bathroom scales oscillates before coming to rest at its final reading. The angular displacement of the pointer is θ radians at time t seconds. The oscillation of the pointer is modelled by the differential equation:

$$\ddot{\theta} + 2\dot{\theta} + 5\theta = 0.3\,.$$

(a) Find the general solution of this differential equation.

(6)

(b) Given that $\theta = 0.2$ and $\dot{\theta} = 0$ when $t = 0$, find the particular solution of the differential equation.

(5)

(c) State, with a reason, the limiting value of θ given by this model.

(2)

(Total 13 marks)

Q26.

The motion of a particle of unit mass, moving along a straight line through O, is governed by the differential equation

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 10x = 325 \sin 3t$$

where x is the displacement from O at time t.

(a) Given that the particle starts from rest at O, solve the equation.

(11)

(b) Describe fully the nature of the motion for large values of t.

(3)

(Total 14 marks)

Q27.

A particle P, of mass m, is suspended from a fixed point O by a light elastic string of natural length $2a$ and modulus $4mn^2a$, where n is a positive constant. The particle is released from rest at a point C, where C is vertically below O and $OC = 2a$.

When the particle is moving with velocity v, it experiences air resistance of magnitude $2mnv$.

(a) The displacement of P below C at time t is x.

Show that x satisfies the equation

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 2n^2x = g$$

(4)

(b) Find x in terms of n, g and t.

(9)

(Total 13 marks)

Q28.

A particle moves along a straight line. At time t, the displacement of the particle is x. Its motion is governed by the differential equation

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} = k \sin 2t$$

where k is a positive constant. The particle starts from rest at the point $x = a$.

(a) Determine x in terms of a, k and t subject to the given initial conditions.

(11)

(b) Find the values between which x oscillates if t is so large that e^{-t} can be neglected.

(4)

(Total 15 marks)

Q29.

A particle P of mass m is suspended from a fixed point O by a light elastic string of natural length a and modulus $6mn^2a$, where n is a positive constant. The particle is released from rest at a point A, where A is vertically below O and distance OA is a.

When the particle is moving with velocity v m s^{-1} it experiences air resistance of magnitude $2mnv$ N.

(a) The displacement of P below A at time t is x.

Show that x satisfies the equation

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 6n^2x = g$$

(3)

(b) Find x in terms of n, g and t.

(10)

(c) Hence show that the first time the particle comes to rest is when $t = \frac{\pi}{\sqrt{5}n}$.

(4)
(Total 17 marks)